Abdelrahman Ehssan

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Education

Alexandria University, Faculty of Science — Alexandria, Egypt
B.Sc. in Computer Science (SIM Program) — *Jun 2026*

Technical Skills

Languages: Python, SQL, Java, C++, C#
Machine Learning & Deep Learning: Supervised Learning, Regression, Classification, Ensemble Methods (Random Forests, Gradient Boosting), SVM, KNN, Feature Engineering, Hyperparameter Tuning, CNNs, Transfer Learning, Object Detection (YOLO), LSTM/GRU, Transformer Fine-Tuning, Sentiment Analysis
LLMs, RAG & AI Agents: LLM Integration, Prompt Engineering, RAG Pipelines, Semantic Chunking, Vector Search, HNSW Indexing, Agentic Workflows, Tool Integration, Workflow Orchestration, Context/State Management
Frameworks & Libraries: PyTorch, TensorFlow, Keras, Hugging Face Transformers, OpenCV, Scikit-learn, FastAPI, LangGraph, LangChain, Cycls SDK
Data & Databases: Pandas, NumPy, PostgreSQL, pgvector, Supabase, SQLGlot, Redis
Cloud, DevOps & Tools: AWS (EC2, S3, RDS), Docker, CI/CD, Celery, MinIO, Git, GitHub

Projects

Multi-Tenant Text-to-SQL & Document Chat Platform (ITI Capstone) — Aug 2026
Python, FastAPI, LangGraph, PostgreSQL/pgvector — *repo*

- Built an end-to-end Applied AI platform that translates natural-language questions into SQL and produces grounded answers across database and document sources, orchestrating LLM reasoning with LangGraph, permission-aware schema exposure, and SQLGlot-validated guardrails.
- Built a RAG pipeline for document question answering – chunking, embedding, and pgvector-based retrieval with reranking across PDF/DOCX/XLSX/CSV/TXT sources – merging SQL and document evidence into one grounded, cited answer.

LearNova – AI Learning Platform (Graduation Project) — Jun 2026
Python, FastAPI, Supabase, PostgreSQL/pgvector, Gemini API — *repo*

- Designed and deployed a full-stack EdTech platform with a RAG-powered AI tutoring engine: scraped and semantically chunked curriculum content into 4,542 metadata-rich chunks, then implemented Supabase/pgvector retrieval with Gemini embeddings and HNSW indexing, achieving 90.0% Precision@3.
- Implemented a dropout-risk prediction model (MAE ∼1.82 days) with Bayesian drift detection, surfaced through admin analytics dashboards and Gemini-generated performance briefings.

Experience

Cycls — Riyadh, Saudi Arabia (Remote)
Applied AI Engineering Intern — *Aug 2026 – Present*

- Building and prototyping production-oriented AI agents using LLMs, RAG, and the Cycls agent SDK for real business workflows, collaborating with the engineering team from concept through iterative test cycles.
- Designing multi-step agent workflows and conversational discovery flows that progressively gather and structure business context into reusable agent state for downstream reasoning.
- Integrating tool and data-source workflows connecting agents to external business systems, while structuring approval-gated execution boundaries and evaluating memory and retrieval strategies for context-aware reasoning.

Information Technology Institute (ITI) — Alexandria, Egypt (Remote)
Applied AI Development Intern — *Jul 2026 – Aug 2026*

- Applied Python, NumPy, Pandas, and object-oriented programming to build modular AI data-processing workflows.
- Built and evaluated computer vision systems: implemented CNNs and OpenCV pipelines for image classification, adapted pretrained ResNet-18 via transfer learning, and applied YOLO for object detection.
- Developed and compared NLP models – from TF-IDF and spaCy embeddings to LSTM/GRU classifiers and a fine-tuned DistilBERT transformer – and applied LLM foundations (tokenization, decoding, prompt engineering) to diagnose grounding failures and design a guarded AI workflow.

Technology Innovation & Entrepreneurship Center (TIEC) — Alexandria, Egypt (Hybrid)
Artificial Intelligence Intern — *Jan 2026 – Feb 2026*

- Developed and evaluated supervised ML models across 5+ structured datasets, benchmarking 6 algorithms (Logistic Regression, Decision Trees, Random Forests, Gradient Boosting, SVM, KNN) for classification and regression – covering feature engineering and hyperparameter tuning – and selected top performers by F1-score, MAE, and R^2, reducing preprocessing time by ∼40% through reusable pipeline scripts.

Certifications

Machine Learning Specialization | DeepLearning.AI & Stanford Online — Jun 2026

Huawei Certified Developer Associate – AI (HCCDA-AI) | Huawei — Mar 2026

Google AI Essentials | Google — Feb 2026